

Master Math for JEE Main & JEE Advanced

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JEE Mains 2020 Jan Chapter wise Question Bank

EM Waves

Q1

Given magnetic field equation is $B = 3 \times 10^{-8} \sin(\omega t + kx + \phi) \hat{j}$, then appropriate equation for electric field (E) will be :

दिये गये चुम्बकीय क्षेत्र की समीकरण $B = 3 \times 10^{-8} \sin(\omega t + kx + \phi) \hat{j}$ है, तब विद्युत क्षेत्र (E) की उचित समीकरण होगी।

- (1) $20 \times 10^{-9} \sin(\omega t + kx + \phi) \hat{k}$ (2) $9 \sin(\omega t + kx + \phi) \hat{k}$
(3) $16 \times 10^{-9} \sin(\omega t + kx + \phi) \hat{k}$ (4) $3 \times 10^{-9} \sin(\omega t + kx + \phi) \hat{k}$

7th Jan Morning

Sol

(2)

$$\frac{E_0}{B_0} = C \text{ (speed of light in vacuum निर्वात में प्रकाश की चाल)}$$

$$E_0 = B_0 C = 3 \times 10^{-8} \times 3 \times 10^8 \\ = 9 \text{ N/C}$$

So इसलिए $E = 9 \sin(\omega t + kx + \phi)$

Q2

If relative permittivity and relative permeability of a medium are 3 and $\frac{4}{3}$ respectively. The critical angle for this medium is.

यदि किसी माध्यम की सापेक्षिक विद्युतशीलता व सापेक्षिक पारगम्यता क्रमशः 3 तथा $\frac{4}{3}$ है तो माध्यम के लिए क्रांतिक कोण होगा।

- (1) 45° (2) 60° (3) 30° (4) 15°

8th Jan Morning

Sol

(3)

$$V = \frac{1}{\sqrt{\mu\epsilon}}$$

$$n = \sqrt{\mu_r \epsilon_r} = 2$$

$$\sin c = \frac{1}{2}$$

$$c = 30^\circ$$

Q3

An EMW is travelling along z-axis.

एक विद्युत चुम्बकीय तरंग **z**- अक्ष के अनुदिश गतिमान है।

$$\vec{B} = 5 \times 10^{-8} \hat{j} \text{ T}$$

$$C = 3 \times 10^8 \text{ m/s}$$

& Frequency of wave is 25 Hz, then electric field in volt/m.

8th Jan Evening

Sol

15

$$\frac{E}{B} = c$$

$$E = B \times c$$

$$= 15 \text{ N/c}$$

Q4

Two electromagnetic waves are moving in free space whose electric field vectors are given by

$\vec{E}_1 = E_0 \hat{j} \cos(kx - \omega t)$ & $\vec{E}_2 = E_0 \hat{k} \cos(ky - \omega t)$. A charge q is moving with velocity $\vec{v} = 0.8 c \hat{j}$. Find the net Lorentz force on this charge at $t = 0$ and when it is at origin.

मुक्त आकाश में दो विद्युत चुम्बकीय तरंगें गति कर रही हैं। जिनके विद्युत क्षेत्र क्रमशः $\vec{E}_1 = E_0 \hat{j} \cos(kx - \omega t)$ और $\vec{E}_2 = E_0 \hat{k} \cos(ky - \omega t)$ के अनुसार दिया गया है। एक आवेश q वेग $\vec{v} = 0.8 c \hat{j}$ से गति कर रहा है। जब यह आवेश पर $t = 0$ तथा मूलबिन्दु पर होता है, तब इस पर कुल लॉरेन्ज बल होगा।

(1) $qE_0(0.4\hat{i} + 0.2\hat{j} + 0.2\hat{k})$

(2) $qE_0(0.8\hat{i} + \hat{j} + 0.2\hat{k})$

(3) $qE_0(0.6\hat{i} + \hat{j} + 0.2\hat{k})$

(4) $qE_0(0.8\hat{i} + \hat{j} + \hat{k})$

9th Jan Morning

Sol

(2)

Magnetic field vectors associated with this electromagnetic wave are given by

विद्युत चुम्बकीय तरंग के संगत चुम्बकीय क्षेत्र सदिश

$$\vec{B}_1 = \frac{E_0}{c} \hat{k} \cos(kx - \omega t) \text{ \& \; } \vec{B}_2 = \frac{E_0}{c} \hat{i} \cos(ky - \omega t)$$

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$$

$$= q(\vec{E}_1 + \vec{E}_2) + q(\vec{v} \times (\vec{B}_1 + \vec{B}_2))$$

by putting the value of $\vec{E}_1$, $\vec{E}_2$, $\vec{B}_1$ & $\vec{B}_2$ के मान रखने पर

The net Lorentz force on the charged particle is आवेशित कण पर कुल लॉरेन्ज बल

$$\vec{F} = qE_0[0.8\cos(kx - \omega t)\hat{i} + \cos(kx - \omega t)\hat{j} + 0.2\cos(ky - \omega t)\hat{k}]$$

at $t = 0$ and at $x = y = 0$

$t = 0$ तथा $x = y = 0$ पर

$$\vec{F} = qE_0[0.8\hat{i} + \hat{j} + 0.2\hat{k}]$$

Q5

An EM wave is travelling in $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$ direction. Axis of polarization of EM wave is found to be $\hat{k}$. Then equation of magnetic field will be

एक विद्युतचुम्बकीय तरंग $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$ सदिश की दिशा में गतिशील है। विद्युतचुम्बकीय तरंग के ध्रुवण का अक्ष $\hat{k}$ के अनुदिश है तो चुम्बकीय क्षेत्र की समीकरण होगी

(1) $\frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos \left[\omega t + k \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) \right]$

(2) $\frac{\hat{i} - \hat{j}}{\sqrt{2}} \cos \left[\omega t - k \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) \right]$

(3) $\frac{\hat{i} - \hat{j}}{\sqrt{2}} \cos \left[\omega t + k \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) \right]$

(4) $\hat{k} \cos \left[\omega t - k \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) \right]$

9th Jan Evening

Sol

(2)

EM wave is in direction विद्युतचुम्बकीय तरंग की दिशा $\rightarrow \frac{\hat{i} + \hat{j}}{\sqrt{2}}$ के अनुदिश है

Electric field is in direction विद्युत क्षेत्र की दिशा $\rightarrow \hat{k}$ के अनुदिश है।

$\vec{E} \times \vec{B} \rightarrow$ direction of propagation of EM wave विद्युतचुम्बकीय तरंग की गति की दिशा

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